Normal Distribution

ANSWER KEY

Properties of the normal distribution

1 State two key properties of the normal distribution curve.

The normal distribution curve is symmetric (bell-shaped) about its mean μ, and the total area under the curve equals 1. (Other valid properties: mean = median = mode; the curve is asymptotic to the horizontal axis.)

2 Explain what it means for the normal distribution curve to be "asymptotic" to the horizontal axis.

It means the curve approaches the horizontal axis as $x \to \pm\infty$ but never actually touches or crosses it — the probability density gets arbitrarily close to zero but is never exactly zero for any finite x.

3 State one similarity and one difference between the normal distribution and the discrete uniform distribution.

Similarity: both are probability distributions in which the total probability sums (or integrates) to 1. Difference: the normal distribution is continuous and bell-shaped, with values clustering near the mean, while the discrete uniform distribution is discrete and every outcome has equal probability.

Standardizing a normal variable

4 A random variable $X \sim N(50, 64)$. Find the z-score corresponding to $x = 62$, using $z = \frac{x - \mu}{\sigma}$.

$\sigma = \sqrt{64} = 8$. $z = \frac{62 - 50}{8} = \frac{12}{8} = 1.5$.

5 A random variable $X \sim N(30, 25)$. Find the z-score corresponding to $x = 23$.

$\sigma = \sqrt{25} = 5$. $z = \frac{23 - 30}{5} = \frac{-7}{5} = -1.4$.

6 A random variable $X \sim N(70, 144)$. Find the z-score corresponding to $x = 88$.

$\sigma = \sqrt{144} = 12$. $z = \frac{88 - 70}{12} = \frac{18}{12} = 1.5$.

Finding probabilities using the standard normal table

7 A random variable $X \sim N(100, 225)$. Find $P(X < 115)$, given that $\Phi(1) \approx 0.8413$.

$\sigma = \sqrt{225} = 15$. $z = \frac{115 - 100}{15} = 1$. $P(X < 115) = \Phi(1) \approx 0.8413$.

8 Using the same distribution $X \sim N(100, 225)$, find $P(X > 130)$, given that $\Phi(2) \approx 0.9772$.

$z = \frac{130 - 100}{15} = 2$. $P(X > 130) = 1 - \Phi(2) \approx 1 - 0.9772 = 0.0228$.

9 A random variable $X \sim N(50, 16)$. Find $P(X < 54)$, given that $\Phi(1) \approx 0.8413$.

$\sigma = \sqrt{16} = 4$. $z = \frac{54 - 50}{4} = 1$. $P(X < 54) = \Phi(1) \approx 0.8413$.

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10 A random variable $X \sim N(80, 100)$. Find $P(X > 95)$, given that $\Phi(1.5) \approx 0.9332$.

$\sigma = \sqrt{100} = 10$. $z = \dfrac{95 - 80}{10} = 1.5$. $P(X > 95) = 1 - \Phi(1.5) \approx 1 - 0.9332 = 0.0668$.

Finding a value given a probability (inverse normal)

11 A random variable $X \sim N(60, 100)$. Find the value x such that $P(X < x) = 0.9332$, given that $\Phi(1.5) \approx 0.9332$.

Since $\Phi(1.5) \approx 0.9332$, the corresponding z-value is 1.5. $x = \mu + z\sigma = 60 + 1.5(10) = 75$.

12 A random variable $X \sim N(80, 64)$. Find the value x such that $P(X < x) = 0.9772$, given that $\Phi(2) \approx 0.9772$.

Since $\Phi(2) \approx 0.9772$, the corresponding z-value is 2. $x = \mu + z\sigma = 80 + 2(8) = 96$.

13 A random variable $X \sim N(45, 49)$. Find the value x such that $P(X < x) = 0.8413$, given that $\Phi(1) \approx 0.8413$.

Since $\Phi(1) \approx 0.8413$, the corresponding z-value is 1. $x = \mu + z\sigma = 45 + 1(7) = 52$.

The probability of a value lying between two bounds

14 A random variable $X \sim N(40, 25)$. Find $P(35 < X < 48)$, given that $\Phi(1) \approx 0.8413$ and $\Phi(1.6) \approx 0.9452$.

$\sigma = 5$. For $x = 35$: $z = \dfrac{35 - 40}{5} = -1$. For $x = 48$: $z = \dfrac{48 - 40}{5} = 1.6$.

$P(35 < X < 48) = \Phi(1.6) - \Phi(-1) = \Phi(1.6) - [1 - \Phi(1)] = 0.9452 - (1 - 0.8413) = 0.9452 - 0.1587 = 0.7865$.

15 A random variable $X \sim N(20, 16)$. Find $P(16 < X < 26)$, given that $\Phi(1) \approx 0.8413$ and $\Phi(1.5) \approx 0.9332$.

$\sigma = 4$. For $x = 16$: $z = \dfrac{16 - 20}{4} = -1$. For $x = 26$: $z = \dfrac{26 - 20}{4} = 1.5$.

$P(16 < X < 26) = \Phi(1.5) - \Phi(-1) = \Phi(1.5) - [1 - \Phi(1)] = 0.9332 - (1 - 0.8413) = 0.9332 - 0.1587 = 0.7745$.

16 A random variable $X \sim N(50, 100)$. Find $P(35 < X < 65)$, given that $\Phi(1.5) \approx 0.9332$.

$\sigma = 10$. For $x = 35$: $z = \dfrac{35 - 50}{10} = -1.5$. For $x = 65$: $z = \dfrac{65 - 50}{10} = 1.5$. Since these bounds are symmetric about the mean:

$P(35 < X < 65) = \Phi(1.5) - \Phi(-1.5) = 2\Phi(1.5) - 1 = 2(0.9332) - 1 = 0.8664$.

The normal approximation to the binomial distribution

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17 A random variable $X \sim B(200, 0.4)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid (both np and $n(1-p)$ should exceed 5).

$np = 200(0.4) = 80$ and $n(1-p) = 200(0.6) = 120$, both exceed 5, so the approximation is valid. Use $X \approx N(80, 48)$, since $\text{Var}(X) = np(1-p) = 200(0.4)(0.6) = 48$.

18 A random variable $X \sim B(150, 0.5)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.

$np = 150(0.5) = 75$ and $n(1-p) = 150(0.5) = 75$, both exceed 5, so the approximation is valid. Use $X \approx N(75, 37.5)$, since $\text{Var}(X) = np(1-p) = 150(0.5)(0.5) = 37.5$.

19 A random variable $X \sim B(100, 0.3)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.

$np = 100(0.3) = 30$ and $n(1-p) = 100(0.7) = 70$, both exceed 5, so the approximation is valid. Use $X \approx N(30, 21)$, since $\text{Var}(X) = np(1-p) = 100(0.3)(0.7) = 21$.

20 A random variable $X \sim B(250, 0.2)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.

$np = 250(0.2) = 50$ and $n(1-p) = 250(0.8) = 200$, both exceed 5, so the approximation is valid. Use $X \approx N(50, 40)$, since $\text{Var}(X) = np(1-p) = 250(0.2)(0.8) = 40$.

Solving for unknown parameters of a normal distribution

21 A random variable $X \sim N(\mu, 36)$ satisfies $P(X < 50) = 0.8413$, given $\Phi(1) \approx 0.8413$. Find μ.

$\sigma = 6$, and the z-value corresponding to probability 0.8413 is $z = 1$. So $\dfrac{50 - \mu}{6} = 1$, giving $\mu = 50 - 6 = 44$.

22 A random variable $X \sim N(\mu, 49)$ satisfies $P(X < 66) = 0.9772$, given $\Phi(2) \approx 0.9772$. Find μ.

$\sigma = 7$, and the z-value corresponding to probability 0.9772 is $z = 2$. So $\dfrac{66 - \mu}{7} = 2$, giving $\mu = 66 - 14 = 52$.

23 A random variable $X \sim N(\mu, 25)$ satisfies $P(X < 40) = 0.9332$, given $\Phi(1.5) \approx 0.9332$. Find μ.

$\sigma = 5$, and the z-value corresponding to probability 0.9332 is $z = 1.5$. So $\dfrac{40 - \mu}{5} = 1.5$, giving $\mu = 40 - 7.5 = 32.5$.

A well-designed word problem: setting a pass mark for an exam

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24 This question has two parts. Scores on a standardized exam are normally distributed with mean $\mu = 68$ and standard deviation $\sigma = 12$. The exam board wants to set a pass mark so that exactly the top 10% of students pass, and is told that the z-value corresponding to the top 10% (i.e. $P(Z > z) = 0.10$) is $z \approx 1.2816$. (a) Find the pass mark, to the nearest whole number. (b) A student scores 90 on the exam. Find their z-score, and state whether they are in the top 10%.

(a) $x = \mu + z\sigma = 68 + 1.2816(12) \approx 68 + 15.38 \approx 83$ (to the nearest whole number). (b) $z = \frac{90 - 68}{12} = \frac{22}{12} \approx 1.833$. Since a score of 90 exceeds the pass mark of approximately 83 (and $z \approx 1.833 > 1.2816$), this student IS in the top 10%.

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